

Case study of the effects of hurricanes on the coupled electricity and water systems of St Kitts

C.A. Johnson & R. Flage

University of Stavanger, Stavanger, Norway

S.D. Guikema

University of Michigan, Michigan, USA

ABSTRACT: When modelling critical infrastructure, it is important to account for the effects of interdependencies. Although there is much literature on how to account for interdependencies, there is a shortage of real-world examples. In an effort to increase the number of real-world examples, a model of the interdependent power and water systems of the Caribbean island of St Kitts has been developed. System dependencies arise due to electrically powered water pumps within the water system. Given the location of the island, it is not uncommon for it to encounter tropical storms, which may result in disruptions to the island's power system. Depending on the severity of the disruption to the power system, the effects may be able to propagate, through the dependencies, into the water system. The developed model uses the track and wind speed of past and simulated hurricanes, or up to date weather predictions, to simulate possible disruptions to the island's power system. Any propagation of these disruptions to the water system through the interdependencies are also simulated. The recent occurrence of hurricane Maria provides a useful case study to compare the output of the coupled system model with the actual effects that resulted due to the hurricane. The models are run with the known track and wind speeds of hurricane Maria. The predicted disruptions to the power system, as well as the cascading effects throughout the water system are then compared to the actual disruptions exhibited in the interdependent systems. The results can be used to validate the model and give an indication if improvements to the current model can be made.

1 INTRODUCTION

When modelling disruptions to critical infrastructure, it is agreed that any interdependencies present need to be taken into account (Buldyrev et al. 2010). There is much literature available on the various methods to model infrastructure interdependencies, however there is a lack of examples which model real interdependent infrastructure systems (Ouyang 2014). Some examples which look at real interdependent systems include Johansson and Hassel (2010) and Dueñas-Osorio et al. (2007).

To provide another real example of interdependent infrastructure modelling, a model of the coupled water and electrical power system on the Caribbean island of St Kitts has been developed by the Guikema Research Group. The model was developed with the intention to see how tropical storms affect the island's coupled power and water systems. The coupled water and power system of St Kitts provides a good opportunity to model a coupled infrastructure system given that they are relatively simple and self-contained as they service only the island of St Kitts.

In the next section, a brief overview of the island of St Kitts and the natural hazards, such as Hurricane Maria, that affect the island will be given. [Section 3](#) will provide more detail about the model of the island's power system, water system and the dependency that exists between the two systems. The results of the model when simulating the effects of Hurricane Maria will be presented in [Section 4](#). This will be followed by a discussion in [Section 5](#) about the challenges associated with developing and validating this model, and more general any interdependent infrastructure model that aims to simulate real systems. The final section, [Section 6](#) will then draw the conclusions of the paper.

2 THE ISLAND OF ST KITTS

St Kitts is the larger of the two islands of the Federation of St Christopher (St Kitts) and Nevis, which are part of the Leeward Island group in the Eastern Caribbean (The Commonwealth, The Official Website of St Kitts and Nevis). The total

population of St Kitts and Nevis is estimated to be just over 50,000. St Kitts is of volcanic origin and thus the centre of the island is mountainous with a highest peak at Mount Liamuiga. Therefore the majority of the population live along the coastline of the island (The Official Website of St Kitts and Nevis).

The total area of St Kitts is 69 square miles or roughly 180 square kilometers and thus is a relatively small island. The small size of St Kitts means that modelling infrastructure systems of the island is feasible, without having to overly simplify the system. These systems are also self-contained due to the geographic constraints that they service only the island of St Kitts.

Due to the location of St Kitts, the natural hazard of tropical storms is an annual threat to the island. The most notable storm to hit St Kitts in recent years was Hurricane George in September 1998, for which the estimated damage costs to St Kitts and Nevis was 445 million United States Dollars (USD) (Relief Web 2002). Other storms that have had great impacts recently on St Kitts include Hurricane Lenny in 1999 with estimated damage costs of around 41 million USD, Hurricane Hugo which hit the island in 1989 and caused damage to roughly 20 percent of the poles in the electricity power system, and Hurricane Luis in 1995 which caused great damage to the power and water systems of the island (Relief Web 1995, Relief Web 1999, US Aid 1990).

2.1 *St Kitts and Hurricane Maria*

The National Oceanic and Atmospheric Administration (NOAA) first issued a tropical storm watch on 16th September 2017, with the first advisory report being issued at 11:00 Atlantic Standard Time (AST). By 17:00 AST, NOAA were recording tropical storm force surface winds of what was then Tropical Storm Maria. At this time a hurricane watch was in effect for St Kitts and Nevis along with Antigua, Barbuda and Montserrat. At the time of Advisory 6 of Maria 17:00 AST, hurricane force winds had been recorded and the storm had been upgraded to a hurricane. The St Kitts hurricane watch had been upgraded to a hurricane warning. As of 19th September 05:00 AST, St Kitts was experiencing tropical storm force winds as Hurricane Maria passed by the south west coast of the island (NOAA). The eye of Hurricane Maria was reported as passing within 90 miles of the island (St Kitts & Nevis Observer 2017). Advisory 15, issued on 19th September at 17:00 AST was the last advisory to issue a hurricane warning to the island of St Kitts (NOAA).

Figure 1 shows the path of the centre of Hurricane Maria as it passed by St Kitts, as well as

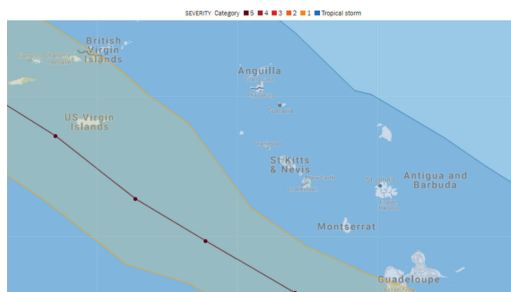


Figure 1. Track of Hurricane Maria—The New York Times, 2017.

the wind speed severity category bands (The New York Times 2017). From Figure 1 it can be seen that St Kitts experienced tropical storm severity wind speeds as Hurricane Maria passed by the south-west of the island.

3 COUPLED POWER AND WATER SYSTEM OF ST KITTS

The island of St Kitts is a relatively small island with an area of 69 square miles. The power and water systems which provide electricity and clean water to the island are self-contained as they do not provide utilities to other islands and can be modelled without having to be over simplified. The water system of St Kitts contains 30 wells which rely on the electricity system to pump the water from the wells into the water system. This dependence can be modelled along with the power and water systems to develop a model of the coupled power and water system for St Kitts.

The water system model was developed using information provided by St Kitts Water Department. The model encompasses the entire clean water distribution system, including the supply sources and demand nodes. Figure 2 shows the modelled water system of St Kitts, including the pipelines, reservoirs and wells. The blue lines in Figure 2 represent the water pipes, the red dots are the water nodes (wells and junctions) and the black nodes are the reservoirs and tanks within the water system.

The water system was modelled using EPANET 2.0, a program that can “perform extended period simulation of hydraulic and water quality behaviour within pressurised pipe networks” EPANET (2000).

The power system was modelled using the limited information available on St Kitts Electricity Company Limited (SKELEC) website. The main lines of the power system were modelled, however

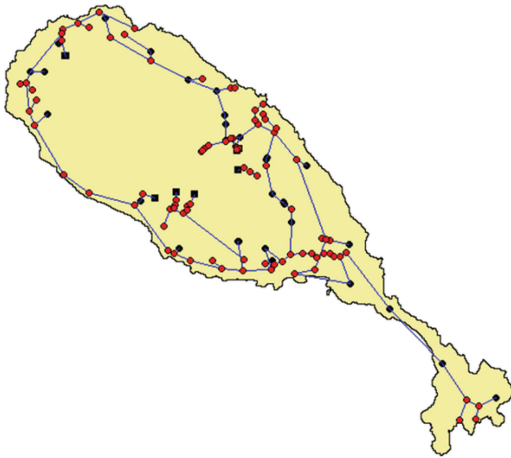


Figure 2. Model of St Kitts water system.

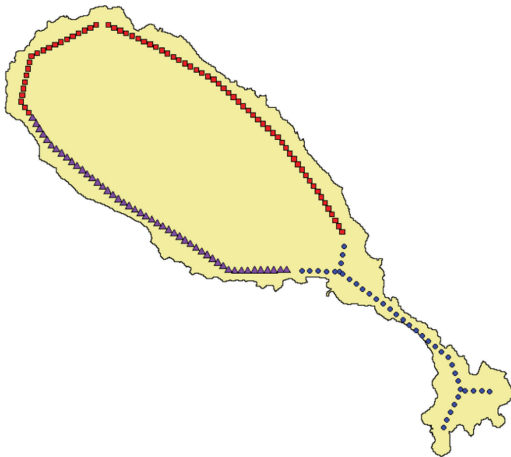


Figure 3. St Kitts electricity power system model.

due to lack of available information the smaller distribution lines of the power system could not be included in the model. As stated in [Section 2](#), most of the island's population resides on the coastal areas of St Kitts. Therefore the 3 main power lines that surround the island are modelled. The model contains 157 nodes, representing 157 wooden electricity poles that the power cables are attached to. The three main power lines can be seen in [Figure 3](#). The northern line is represented by red squares and contains 70 poles, the southern line is shown as blue dots and contains 39 poles and the western line is shown as purple triangles and contains 48 of the 157 poles.

For each pole, a fragility curve giving the probability of failure given the on-island wind speed

has been determined based on historical damage reports of hurricanes that were readily available to the public. If one pole fails in a power line, then all the poles downstream of the failed pole will also be modelled as inoperable.

The water system depends on the power system at each of the 30 wells within the system to get the water from the well into the water distribution system. To model the water system's dependency on the power system, each well is dependent on the closest electricity node, or pole, to provide power to the well. During the simulation, if a node becomes inoperable that a well depends on, the well that depends on it is then removed from the water system model. [Figure 4](#) shows the modelled electricity lines, as black triangles, and the water wells as red squares and other water nodes as blue dots. Sixteen wells depend on nodes within the northern power line, 7 wells depend on electricity from the western line and 8 wells depend on the southern line.

The track as well as the maximum wind speed recorded every 6 hours for the centre of Hurricane Maria from 17th to 28th September 2017 was used to estimate the maximum speed of the hurricane over St Kitts using a model previously developed by Guikema et al. (2014). This provided the probability of failure for each of the 157 poles within the electricity power system. These probabilities were then used in the MATLAB model developed within the Guikema Research. To look at the effects of Hurricane Maria on St Kitts coupled power and water systems, the MATLAB model was run for 60000 iterations, where first it was modelled which poles would fail or break due to the hurricane before modelling the cascading effects of these poles down each power line and through to the water system.



Figure 4. Coupled power and water system of St Kitts.

4 RESULTS

4.1 Electricity power system

For each of the 60000 iterations that the model completes, each pole is recorded as failing due to the wind or not. Therefore, the model gives the frequency of initial pole failure, which can be seen in Figure 5. The frequency, given as a percentage, of each pole failing due to Hurricane Maria is shown using the colour scale as shown in Figure 5.

There are a few poles with relatively low frequency of failing in the southern line, closest to the centre of the island. However, the southern line is mostly unaffected by Hurricane Maria. On the northern power line there is one pole, around a third of the way along the line that has a high frequency of failure. Failure of this pole would cause disruptions to the rest of the line up to the north-east of the island, and thus any wells that depend on any nodes on this section of the northern line to also be disrupted. On the western power line there are more poles within the line that have initial failures due to the hurricane. This is expected as Hurricane Maria passed by the south-west of the island. The most frequent initial failure in the western line is a pole that lies around halfway along the western coast. This would cause disruptions along the rest of the line, all along the western coastline, and these disruptions would cascade to any wells that depend on this section of the western power line.

4.2 Actual power outages

When looking for data of what affects Hurricane Maria had on the electricity system of St Kitts, the only information found came from SKELEC website. This information was a restoration update that named residential areas that were experiencing power outages due to Hurricane Maria. The press release stated “Most of our feeders remained intact and online during the passage of the storm. Some like the Canada feeder which services Conaree, Halfmoon and Canada Estates came offline, also Basseterre North Buckley’s to Trinity also is fully offline” SKELEC (2017).

Figure 6 shows the areas that SKELEC reported to have experienced outages due to Hurricane Maria. The three areas to the north of the island are Canada, Halfmoon and Conaree. Canada has been filled red rather than just outlined as the Canada feeder was the feeder identified as causing these outages to the north of the island. These three outlines were not specified by SKELEC but are the outline of the three areas named by SKELEC in the press release. Due to no further information being found in relation to power outages due to the hurricane this is best representa-

tion of possible affected areas that can be given. The larger section on the south-west of the island represents the area from Basseterre North Buckley’s to Trinity. The outline encases all residential areas between, and including, Basseterre North Buckley’s to Trinity. Again, the press release from SKELEC is the only information on which to base the actual areas affected and thus the areas cannot be more specific.

Figure 7 shows the results of the model for frequency of initial pole failure compared to the areas that SKELEC reported to experience power outages due to Hurricane Maria. When looking at the predictions and actual areas to exhibit outages on the northern power line, the high frequency of a

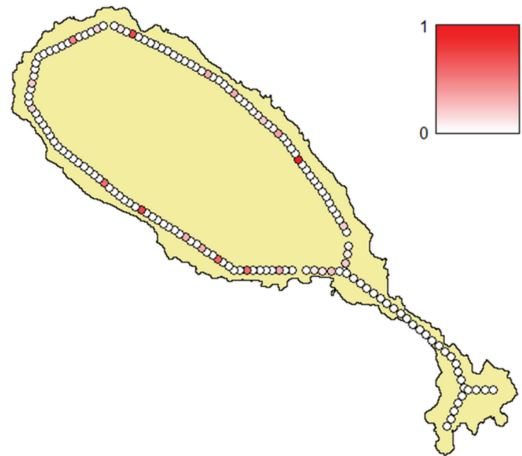


Figure 5. Frequency of initial pole failure.



Figure 6. Areas of St Kitts that experienced power outages due to Hurricane Maria.

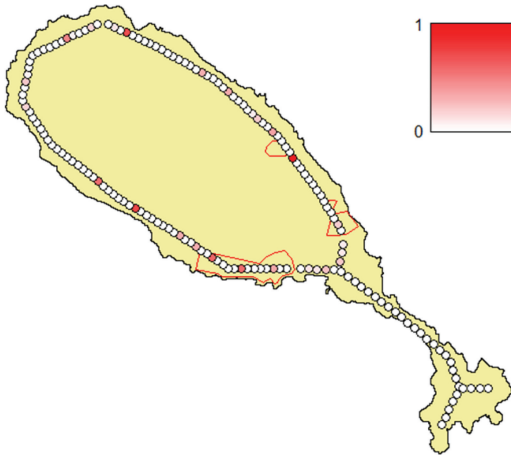


Figure 7. Frequency of initial pole failure compared to areas to experience power outages.

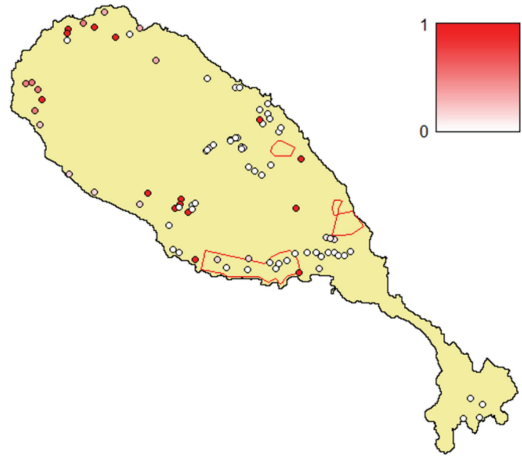


Figure 8. Frequency of water nodes with pressure lower than 20 psi.

pole break on the northern side of the island given by the model coincides with the press release that the northern side was affected such that a feeder came offline. When looking at the south-west area that experienced outages, if the main line was affected by the hurricane, it would understandable to see more outages along the line. However, these outages could have been caused to failures or disruptions within the smaller distribution lines that are not included in the model.

4.3 Water system

When looking at disruptions to the water system, the frequency that nodes exhibit pressure below 20 psi as well as 0 psi are both considered. The pressure of the nodes within the water system is important to measure as if the pressure becomes negative this can cause the water to flow in the wrong direction, or cause pipes to break. These both can cause the water to become contaminated. Pressures lower than 20 psi are investigated as in the USA this is the minimum pressure required to use a water system for firefighting purposes (EPA 1992).

Figure 8 shows the frequency that water nodes exhibited pressures lower than 20 psi. Comparing Figure 8 to Figure 5 the low pressures towards the lower end of the island coincide to the areas in which the initial pole failures occurred. The low pressures exhibited towards the north-western part of the island are due to the initial failures in both the northern and western power lines cascading to the north of the island.

Figure 9 shows the frequency that the water nodes exhibited pressures less than 0 psi. There are

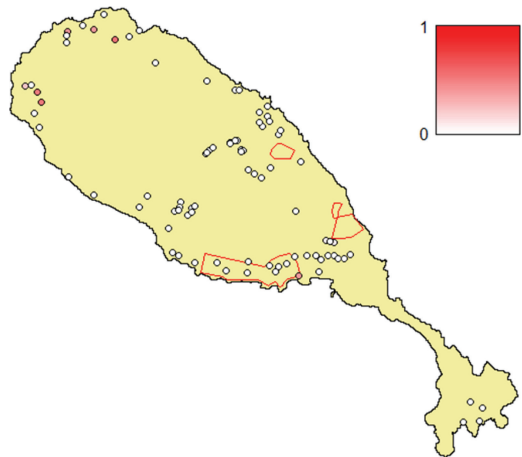


Figure 9. Frequency of water nodes with pressure lower than 0 psi.

few instances of nodes exhibiting pressure lower than 0 psi, with a few occurring on towards the north-west end of the island. Although the initial aim was to compare the results of the model to how Hurricane Maria affected the coupled infrastructure power and water system of the island, it was challenging to find any data related to disruptions to the water system.

The location of wells compared to the actual outage data was also considered to see if any wells would have been without power during the outages.

When looking at Figure 10, there are three wells that are situated in the south-west area of

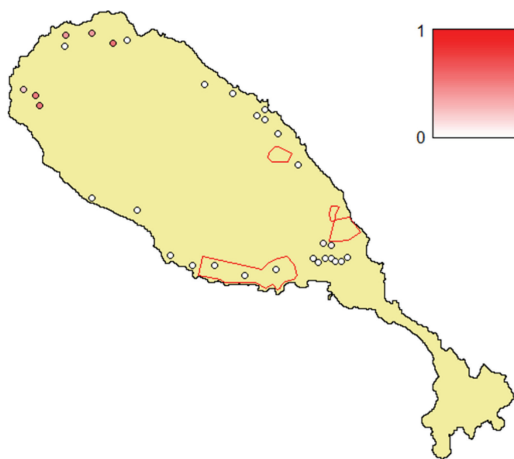


Figure 10. Frequency of water wells with pressure lower than 0 psi and reported power outages.

St Kitts that experienced power outages. To further test the water system model, these three wells were removed from the model, as is they had lost power. However, all other nodes in the water system model maintained a pressure above 0 psi, thus even if these three wells were without power, the water system would not have any disruptions.

Although no information directly related to disruptions to the water system was found, the Prime Minister of St Kitts and Nevis stated that the island's infrastructure such as the electricity and water systems "sustained extensive damage" (Relief Web 2017).

5 CHALLENGES WHEN MODELLING AND VALIDATING REAL INFRASTRUCTURE MODELS

When modelling the coupled power and water infrastructure system of St Kitts, information regarding the water system was provided by St Kitts Water Department. This, along with ongoing communication with the water department while developing the model, allowed an extensive model of the system to be developed. However, when developing the electricity model, the only information available was that in the public domain, primarily on SKELEC website. This meant that the model was an oversimplified representation of the main power lines in the island's electricity system.

Accessing and collecting data to form the basis of infrastructure models is a well-documented issue with constructing models of independent infrastructures as well as interdependent infra-

structures (Ouyang 2014, Johansson and Hassel 2010, Rinaldi et al. 2001). Another problem highlighted by Ouyang (2014) is that infrastructure are large, complex systems that are constantly changing and thus the data used to produce the model may not be relevant for very long after the model has been developed. Any major changes to the infrastructure would have to be updated within the model.

When trying to validate the model, information on the disruptions to both the water and power systems due to Hurricane Maria were also hard acquire. The electricity company did have some information available, however, this is aimed at the customers of the infrastructure and is there to reassure them that the company is aware of the problems and are fixing the issues. The information relates only to general areas, and not specific parts of the system that would be useful when validating the model.

Within the USA, publicly available outage data of power companies is becoming increasingly more available. Most power utility companies are now showing daily outage data on their websites, see the website Power Outage for more information on which utilities provide outage data. However, other utilities in the USA, such as those who manage water infrastructure are still less willing to share outage data.

6 CONCLUSION

The aim of the paper was to compare the results of the coupled power and water system of St Kitts to the disruptions the systems exhibited during Hurricane Maria. Although the electricity model is a simplified representation of only the island's main power lines, the areas that experienced power outages did coincide with poles that had a high frequency of initial failure. This showed that the model is quite accurate on estimating the maximum wind speed for the different areas of the island. However, as the smaller distribution lines could not be included in the model and limited outage information is available, it is difficult to access the accuracy of the model. Improvements could be made by increasing the complexity of the model, if the necessary information can be obtained from St Kitts Electricity Company.

Although the water system model is a good representation if the island's water system, the lack of data available relating to disruptions due to Hurricane Maria again means it is difficult to validate the model. The lack of available information relating to infrastructure systems is a well-recognised challenge when developing models of infrastructure systems.

REFERENCES

- Buldyrev, S.V., Parshani, R., Paul, G., Stanley, H.E. & Havlin, S. 2010. Catastrophic cascade of failures in interdependent networks. *Nature*, 464, 1025.
- Dueñas-Osorio, L., Craig, J.I., Goodno, B.J. & Der Kiureghian, A. 2007. Seismic response of critical interdependent networks. *Earthquake Engineering & Structural Dynamics*, 36, 285–306.
- EPA, U. 1992. Manual of Small Public Water Supply Systems.
- EPANET 2000. [Online] Epanet 2 Users Manual Available at: <https://nepis.epa.gov/Adobe/PDF/P1007WWU.pdf> [Accessed on 21 December 2017].
- Guikema, S.D., Nateghi, R., Quiring, S.M., Staid, A., Reilly, A.C. & Gao, M. 2014. Predicting Hurricane Power Outages to Support Storm Response Planning. *Access, IEEE*, 2, 1364–1373.
- Johansson, J. & Hassel, H. 2010. An approach for modeling interdependent infrastructures in the context of vulnerability analysis. *Reliability Engineering and System Safety*, 95, 1335–1344.
- NOAA National Oceanic and Atmospheric Administration. [Online] Hurricane Maria Advisory Archive Available at: <http://www.nhc.noaa.gov/archive/2017/MARIA.shtml?> [Accessed on 21 December 2017].
- Ouyang, M. 2014. Review on modeling and simulation of interdependent critical infrastructure systems. *Reliability Engineering and System Safety*, 121, 43–60.
- Relief Web 1995. [Online] Caribbean—Hurricane Luis Sep 1995 UN DHA Situation Reports-10 Available at: <https://reliefweb.int/report/antigua-and-barbuda/caribbean-hurricane-luis-sep-1995-un-dha-situation-reports-1-10> [Accessed on 21 December 2017].
- Relief Web 1999. [Online] Hurricane Lenny OCHA Situation Report No. 7 Available at: <https://reliefweb.int/report/anguilla/hurricane-lenny-ocha-situation-report-no-7> [Accessed on 21 December 2017].
- Relief Web 2002. [Online] Caribben—Hurricane George Appeal No. 29/1998 final report. Available at: <https://reliefweb.int/report/antigua-and-barbuda/caribbean-hurricane-georges-appeal-no-291998-final-report> [Accessed on 21 December 2017].
- Relief Web 2017. [Online] Prime Minister Dr. the Hon. Timothy Harris' Post-Hurricane Maria Address Available at: <https://reliefweb.int/report/saint-kitts-and-nevis/prime-minister-dr-hon-timothy-harris-post-hurricane-maria-address> [Accessed on 21 December 2017].
- Rinaldi, S.M., Peerenboom, J.P. & Kelly, T.K. 2001. Identifying, understanding, and analyzing critical infrastructure interdependencies. *Control Systems, IEEE*, 21, 11–25.
- Skelec 2017. [Online] Hurricane Maria Restoration Update Available at: <http://www.skelec.kn/hurricane-maria-restoration-update-wednesday-september-20-2017/> [Accessed on 21 December 2017].
- St Kitts & Nevis Observer 2017. [Online] Hurricane Maria remains powerful, dangerous hurricane Available at: <http://www.thestkittsnevisobserver.com/breaking-news/hurricane-maria-remains-powerful-dangerous-hurricane-news-update/> [Accessed on 21 December 2017].
- The Commonwealth [Online] St Kitts and Nevis Available at: <http://thecommonwealth.org/our-member-countries/st-kitts-and-nevis> [Accessed on 21 December 2017].
- The New York Times 2017. [Online] Maps: Hurricane Maria's Path Across Puerto Rico Available at: [Accessed on 21 December 2017].
- The Official Website of St Kitts and Nevis [Online] About St Kitts and Nevis Available at: <https://www.gov.kn/> [Accessed on 21 December 2017].
- US AID 1990. [Online] After-action Report of the Hurricane Hugo OFDA Disaster Relief Team Available at: http://pdf.usaid.gov/pdf_docs/Pnabg072.pdf [Accessed on 21 December 2017].